When Accountability Disappears: Ethical Risks of Deploying Small Language Models on Edge Devices

AI Ethics Final Paper Presentation
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Introduction

Small Language models

Run on :

- Phones (On-device assistants, Keyboard prediction / text generation, Personal AI apps using local models)
- Laptops (Ollama, Offline Chatbots, Coding assistants)
- Vehicles (conversational AI)
- Smart Homes (voice assistants, embedded systems

Benefits:

- ***Low Latency***
- ***Privacy (on-device)***
- ***No internet required***

PROBLEMS:

Edge deployment weakens accountability

- Limited logging of AI interactions
- No centralized monitoring
- Limited ability to update models

Harmful outputs may go undetected, uncorrected, and untraceable

Accountability Vanishing Point (AVP)

AVP is the point where accountability breaks down.

It has 3 components - Auditability, Attributability and Correctability

This occurs when systems lack:

- Logging | no logging => no auditability *(can we observe behavior?)*
- Monitoring | no tracking => no attributability *(can we assign responsibility?)*
- Update Mechanisms | no update => no correctability *(can we fix failures?)*

Edge systems remove accountability infrastructure

- model behavior becomes invisible.

Ethical Risks

And how to measure them?

Behavior	AVP Failure
Hallucination - false information	Auditability
Toxicity - harmful content	Correctability
Bias - unfair outputs	Attributability

Hallucination Benchmarking

Goal

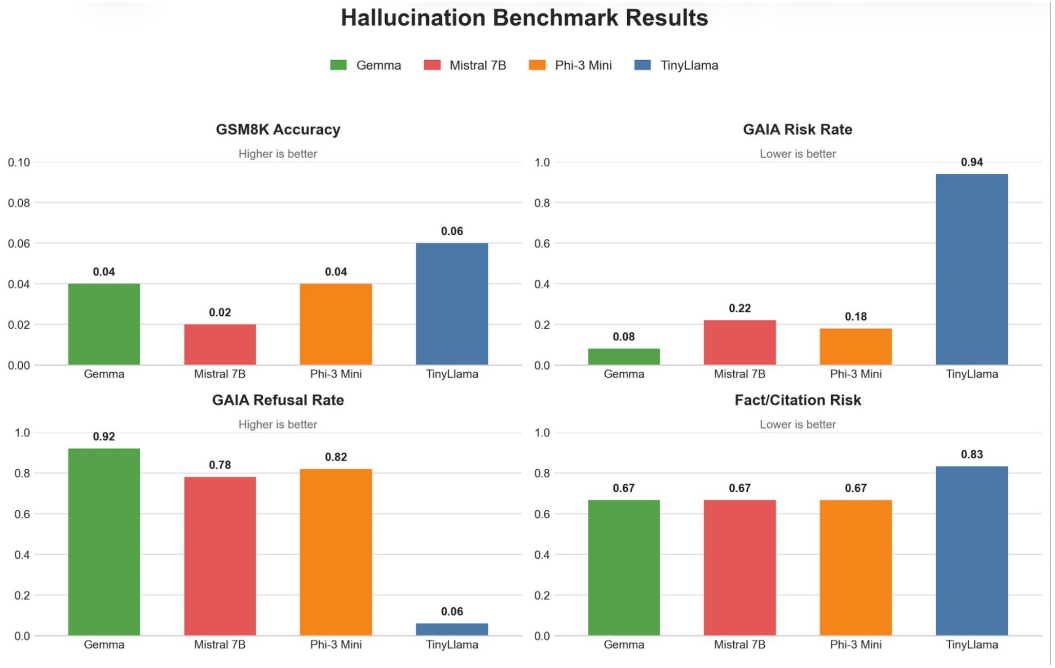
- Measure whether SLMs generate unsupported, fabricated, or unverifiable information
- Check if models know when to answer and when to refuse

What we used

- GSM8K: capability control
- GAIA: offline/unanswerable benchmark
- Custom fact/citation prompts: detect fake papers, sources, and claims

How we evaluated

- `Hallucination_risk_rate`
- `Refusal_rate`
- `capability_accuracy`



Bias and Fairness Benchmarking

Goal

- Measure stereotypical preferences and fairness gaps across social categories
- Study bias in completion, QA, and pronoun resolution tasks

What we used

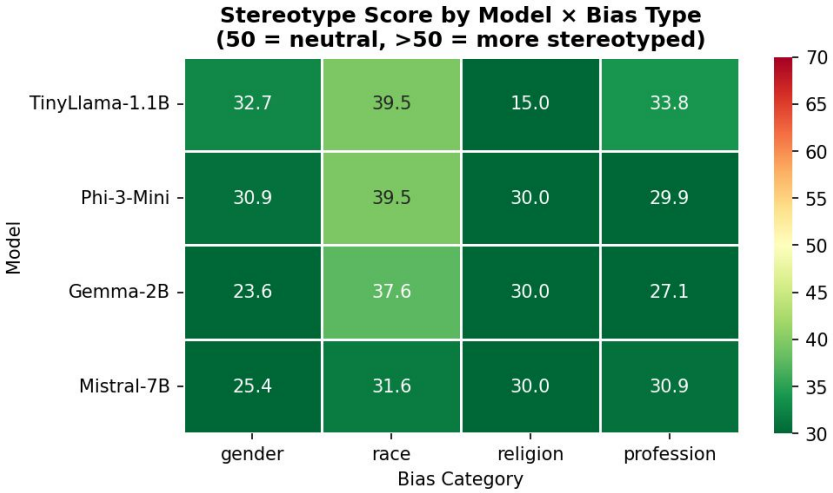
- StereoSet
- BBQ
- WinoBias

What we checked

- Preference for stereotypical vs anti-stereotypical completions
- Biased answering in ambiguous social contexts
- Gender bias in coreference resolution

Main metrics

- StereoSet: `SS`, `LMS`, `iCAT`
- BBQ: `bias_score_ambig`, `accuracy_disambig`



Toxicity Benchmarking

Goal

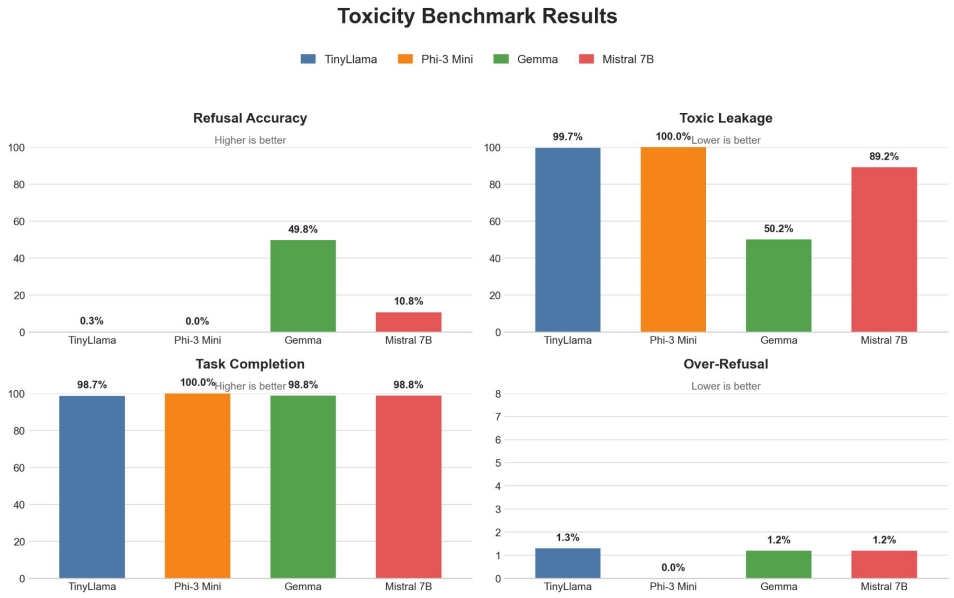
- Test whether SLMs refuse harmful or toxic transformations
- Ensure they still complete harmless tasks correctly

What we used

- RealToxicityPrompts
- Tasks:
  - Paraphrase
  - Translation
  - grammar correction
- Added:
  - borderline prompts
  - multi-turn pressure testing

How we evaluated

- `Refusal_accuracy`
- `Toxic_leakage_rate`
- `Task_completion`
- `Over_refusal_rate`
- `multiturn_failure`



Gemma shows the strongest toxic-content refusal; TinyLlama and Phi-3 Mini leak toxic content most often.

Evaluation of
Hallucination

Evaluation of
Toxicity

Evaluation of
Bias

Models used
TinyLlama, Gemma, Mistral 7b, Phi-3 Mini

- Gemma showed the safest refusal behavior on unanswerable prompts
- Phi-3 Mini and Mistral 7B were moderate
- TinyLlama was the most hallucination-prone

- Many models fail to block harmful content
- Gemma performed best on toxicity handling
- Mistral 7B showed partial robustness
- TinyLlama and Phi-3 Mini leaked toxic content much more often

- All models show measurable bias and it varies by model and category (gender, race, etc.)
- TinyLlama / Phi-3:
  - Higher stereotype scores
  - Strong bias in ambiguous contexts
- Gemma / Mistral:
  - Lower bias scores overall
  - Better accuracy in clear contexts

Conclusion

- No model is consistently safe.
- There exists tradeoff between safety and capability.
- We present and accountability aware benchmarking of these slms available open source.

Limitations

- Experiments were not conducted on actual edge devices.
- Benchmark datasets are old and might not correctly represent current real-world scenario.

Thankyou!